

SHILUVELO MAKHUBELE

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SOFTWARE ENGINEER

I am a passionate and driven computer science graduate currently doing my honours degree specializing in software engineering, Computer security, and digital forensics. With hands on experience in both full-stack development and cybersecurity, I have developed a strong foundation in cloud computing, forensic readiness in IoT, and secure software design. My expertise extends to modern technologies such as React, cloud-based security, and digital forensic tools, which I apply to innovative solutions that prioritize security and user experience. I have a keen interest in developing secure, efficient, and visually appealing applications, whether through my ongoing digital forensic prototype for Apple's HomeKit ecosystem or my entrepreneurial venture building an e-commerce website. My goal is to contribute to impactful projects that align with my technical background and problem-solving capabilities.

EDUCATION

BscHons in Computer Science

Majors: Software Engineering, Computer Security, Digital Forensics and Image processing, Software Education
University of Pretoria-2025

Bsc Computer Science

Majors: Listed in academic record
University of Pretoria-2023

PROFESSIONAL EXPERIENCE AND PROJECTS

GIZ

Feb 2025 - present

Stationed at South African Weather Services

Database Administrator Intern

- Contributed to major data recovery efforts following a severe cyber-attack, restoring data for 14 national weather stations and rebuilding critical databases.
- Collaborated with senior DBAs and IT specialists to recover data from backups and repair damaged systems.
- Led modernization of legacy Java applications that process real-time and aggregated weather station data, achieving 90% restoration across affected stations.
- Initiated and contributed to the development of a web-based tool using React.js (frontend) and Node.js (backend) to simplify data management tasks for non-technical staff.
- Gained practical experience in database management (SQL, stored procedures), Java development, cybersecurity, disaster recovery, and agile methodologies.

NVS System Update

Jan 2024 - Mar 2024

Full stack Developer

- With a group of 4 including a senior developer we updated the NVS system, a provider of pre-employment and background screening solutions.
- Skills: Worked with React, .NET, and DevOps pipelines.

Graduate Trainee

- Participated as part of a bursary program to gain industry experience
- Skills:
 - Adaptability: Quickly learned processes and adjusted to new environments
 - Time Management: Prioritized tasks efficiently to meet deadlines
 - Communication, and teamwork

PROJECTS

Honours Research Project

Feb 2024 - Oct 2024

Computer Science Honours Research

- focuses on digital forensic readiness for Internet of Things (IoT) devices, specifically examining Apple's HomeKit ecosystem. This study develops techniques for gathering, securing, and analyzing digital evidence from IoT devices, with an emphasis on pre-incident collection using the prototype tool "SecureCollect" which I developed.
- Skills: React & Node.js, MongoDB, Python, Wireshark, Flask, API development

COS720 Project

Feb 2024 - Jul 2024

Computer Security Honours Project

- I Built a cloud-based application with security features that showcased my understanding of cloud-security principles, I used AWS's EC2 instance as my cloud environment.
- Skills: AWS EC2,React,Node.js,DynamoDB,Logging,Monitoring,SSL/TLS setup AWS inspector,Encyption and hashing.

Capstone Project

Jan 2023 - Oct 2023

Final year project

- We Built an Event Organizer app for a client(EPI-USE) a part of our final year computer science project.
- Skills : Full-stack development using Angular, Nodejs ,Git/Github, MongoDB, AWS RESTful API creation ,JWT, MongoDB data modelling ,Event scheduling and user management ,Unit and integration testing.
- Application of the Software Development Lifecycle and Agile software development methodology,
- Application of Software architectures Patterns (Microservices, Client-Server,Service-Oriented Architecture, etc.)

KEY COMPETENCIES

Programming Languages:

- Java
- C++
- Python
- Javascript
- SQL

Frameworks:

- Angular/Ionic
- React js
- Springboot

Other tools and skills:

- Agile Methodologies
- Algorithms and data structures
- NodeJS
- Database Management
- Apis
- Aws ec2
- Ci/cd pipelines
- Git/Github
- Unit ,Integration and end-to-end testing
- Design patterns

Soft Skills:

- Teamwork
- Problem Solving
- Critical Thinking
- Dependable
- Time management
- Adaptability
- Ability to work under pressure

LANGUAGES

- English- Proficient
- Xitsonga- Native

REFERENCES

- Lefa Morabe - +27781673677- Systems Developer (SAWS)
- Bawinile Mgedesi - +27829267151- Senior Database Admin(SAWS)
- Mark Makura - +27846062447- Research supervisor(UP)